PCRQ 8

Graded

Student

VK Nguyen

Total Points

3 / 4 pts

Question 1

(no title) **1** / 1 pt

✔ **+ 1 pt** Correct

+ 0 pts Incorrect

Question 2

(no title) **0** / 1 pt

+ 1 pt Correct

✔ **+ 0 pts** Incorrect

Question 3

(no title) **1** / 1 pt

✔ **+ 1 pt** good faith answer

+ 0 pts Blank

Question 4

(no title) **1** / 1 pt

✔ **+ 1 pt** good-faith answer

+ 0 pts Blank

this means that if q was negative, -q would be positive, and you could do the outlines proof in the notes for -q to then show it for q.

Q1
1 Point

Let S be a nonempty subset of real numbers, which is bounded above. Which statement is correct?

- ○ Every upper bound of the set is less than its supremum.
- ◉ A real number is an upper bound of the set if and only if it is greater than or equal to the supremum.
- ○ The supremum is the only upper bound of the set.
- ○ Every real number strictly less than or equal to the supremum belongs to the set.

Q2
1 Point

Let S be a nonempty subset of a totally ordered set. Which statements are true? Select all that apply.

- ☐ The set S may have a supremum but no maximum.
- ☑ The set S may have a maximum but no supremum.
- ☐ If S has a supremum, then $\sup(S) \in S$.
- ☑ If S has a maximum, then $\sup(S) = \max(S)$.

Q3
1 Point

As you prepare for the midterm, which concepts or types of problems have you found most confusing so far? Please be as specific as possible about what you find difficult.

inverse function, inverse composition of functions, inverse image, pre image

I feel like I don't have a good understanding about its definition, and I struggles with proving to inverse functions equal.

Q4

1 Point

Please write briefly (in two or three sentences) what single point of the notes of Lecture 9 you found to be the most difficult or confusing. If you did not find any part of it difficult or confusing, please say what part(s) you found most interesting.

Theorem 2.1. is quite difficult to follow for me. I don't understand the part about replacing q with -q, if necessary, we may assume that q>0.